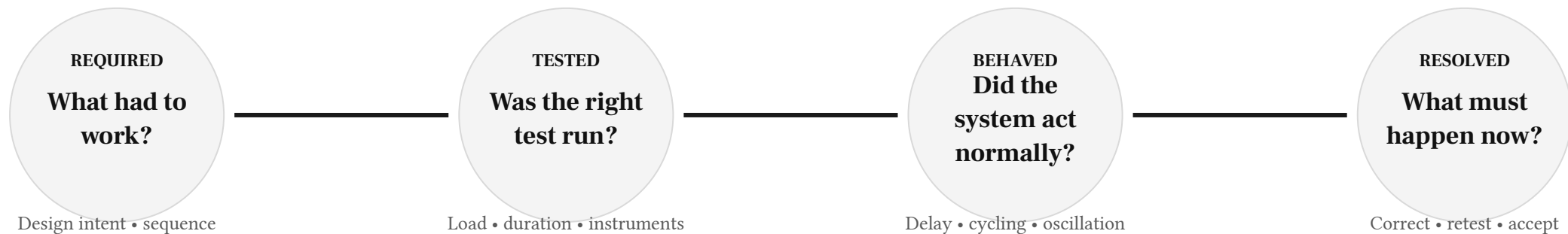


Prove that the facility works—not merely that the test paperwork is complete

We make sure the data centre did not merely complete its commissioning paperwork—it actually passed the right tests and behaved as it was supposed to.



Know what had to work.

Prove it was tested.

Detect abnormal behaviour.

Fix it while recovery is available.

For owners, commissioning managers and receiving operations teams deciding whether to accept, retest, correct or escalate.

The report says ‘pass’—but the raw behaviour tells a different story

INTEGRATED UTILITY-FAILURE TEST

All cooling units restart and the commissioning report records a pass.

WHAT THE RAW TRENDS SHOW

60% of the required thermal load was reached

✚3 min CDU-14 restarted later than its peers

11 min two control valves continued to oscillate

WHAT THE FORM SAYS

PASS

Utility-failure sequence complete

WHAT THE EVIDENCE SAYS

UNPROVEN

The required condition was not reached, and abnormal recovery needs engineering review.

THE DECISION

Correct the controls and repeat the scenario at contractual load before acceptance.

Completed paperwork is not the same as valid evidence; a threshold pass is not the same as normal behaviour.

A cited acceptance decision pack—not another completion dashboard

UNPROVEN	
Cooling recovery requirement CXR-042 is not proven.	
Test condition	IST-07 reached only 60% of required load
Peer behaviour	CDU-14 restarted 174 seconds after the median
Control stability	Valves V-22 and V-24 oscillated for 11 minutes
Required action	Correct controls; retest at contractual load
Decision owner	Commissioning Manager before provisional acceptance

ASSURANCE STATUS

●	Proven	Valid evidence supports it
●	Failed	A criterion was not met
●	Unproven	Test or evidence is insufficient
●	Anomalous	Behaviour requires review
●	Exception	Authorised departure
●	Cannot determine	Identity or evidence is weak

The product prepares the evidence, abnormal behaviour and retest priority. The CxA, engineer and owner retain acceptance authority.

Find the defect while the delivery team can still correct it

TODAY

Teams verify that tests and sign-offs exist under closeout pressure.

- A completed form can hide a weak test
- Evidence is scattered across systems
- Static thresholds miss subtle defects
- All open items compete for attention
- Commissioning knowledge stops at handover

WITH COMMISSIONING ASSURANCE

Acceptance is tied to valid evidence and actual system behaviour.

- ✓ Check load, duration, sequence, instruments and interruptions
- ✓ Link requirement, script, trend, issue, retest and sign-off
- ✓ Find delay, cycling, oscillation and peer outliers
- ✓ Rank correction and retest by consequence and deadline
- ✓ Carry a targeted watchlist into early operation

Practical value: faster evidence review, targeted retests, earlier defect correction, clearer acceptance and protected recovery rights.

Connect the requirement to the test, raw evidence, acceptance and early operation

THE DATA WE NEED

1	Required behaviour	OPR • design basis • specs • sequences
2	Delivered configuration	Submittals • deviations • settings
3	Test method	Scripts • prerequisites • criteria
4	Recorded proof	Raw BMS/EPMS trends • instruments
5	Decision history	Issues • retests • waivers • sign-offs
6	Early operation	Alarms • work orders • warranty dates

HOW THE ANSWER IS BUILT

- 1 Establish what must be proven**
Resolve the current requirement and installed state
- 2 Test whether the evidence is valid**
Check identity, load, duration, sequence and measurement
- 3 Reconstruct actual behaviour**
Align commands, states and trends; find abnormal patterns
- 4 Prepare the acceptance action**
Rank correction, retest, exception and recovery deadlines

The defensible asset is the golden thread: requirement → test → evidence → acceptance → early operation.

What must be true before we test the use case

Start with one critical system and a bounded set of initial tests and retests—not the entire commissioning programme.

✓	Current requirements	Approved sequence, criteria and delivered configuration
✓	Raw test trends	High-resolution evidence for the full test window
✓	Reliable identity	Assets, point tags, test steps and issues can be linked
✓	Known decisions	Initial attempts, retests, dispositions and acceptance history
✓	Qualified reviewers	FivD specialists can validate material findings

If raw trends or credible asset identities are unavailable, position this as evidence automation—not defect-prevention intelligence.

Plain language on the left; technical capability on the right

WHAT THE CLIENT HEARS	WHAT WE BUILD	WHAT IT DOES
 Read requirements and test records	Document intelligence	Extract criteria, steps, results, exceptions and citations
 Know which asset and point is which	Entity resolution	Link assets, sensors, tests, issues and revisions
 Check whether the test was valid	Deterministic rules	Assess prerequisites, load, duration, sequence and calibration
 Replay what actually happened	Temporal processing	Align commands, states, readings, overrides and test steps
 Find behaviour worth investigating	Anomaly detection	Detect peer outliers, oscillation, delay, cycling and drift
 Route the right acceptance action	Governed agents	Assemble evidence and route correction, retest or exception

Control boundary: no autonomous certification, acceptance, control-system change, OT write-back or commercial exception.